PROJECT REPORT

Customer Churn Prediction

NIKHIL TIWARY

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1.Introduction

Customer churn refers to the situation when customers stop using a company’s services. It is a major challenge for businesses, especially in the telecom industry, where competition is very high.

Predicting customer churn helps companies identify customers who are likely to leave. This allows businesses to take preventive actions such as providing better offers, improving services, or engaging customers more effectively.

In this project, machine learning techniques are used to analyze customer data and predict whether a customer will churn or not.

2.Problem Statement

In today's competitive business world, it's important to keep customers happy so they don't stop using our products or services. We want to develop a model that can predict which customers are likely to stop using our service, so we can take steps to keep them.

Customer churn can lead to a loss of revenue and a decrease in customers. We want to use machine learning to build a model that can accurately predict which customers are likely to churn based on their past behaviour, demographics, and subscription details. This will help us target high-risk customers with personalized retention strategies.

We want to create a solution that will help us keep customers happy and using our products or services for the long term.

3.Objective

- **Predict churn: Build a supervised classifier to predict Churn=Yes vs No.**
- **Data understanding: Summarize and preprocess the Telco dataset (7,043 rows, 21 columns).**
- **Key drivers: Identify which features most influence churn (e.g. contract, tenure, charges).**
- **Model performance: Evaluate using accuracy, precision, recall, F1 and confusion matrix.**
- **Business insight: Provide actionable recommendations to reduce churn based on model findings.**

4.Data Description

Dataset The dataset used in this project is the Telco Customer Churn Dataset.

Total Records: 7043 customers

After preprocessing: 7032 records

Number of features: around 30

Key Features:

Customer Details: Gender, Senior Citizen

Service Details: Internet service, Online security, Tech support

Account Details: Tenure, Contract type

Billing Details: Monthly charges, Total charges

Target Variable:

Churn: Indicates whether a customer leaves the service (Yes/No)

Dataset consists customer information for a customer churn prediction problem. It includes the following columns:

CustomerID: Unique identifier for each customer.

Name: Name of the customer.

Age: Age of the customer.

Gender: Gender of the customer (Male or Female).

Location: Location where the customer is based, with options including Houston, Los Angeles, Miami, Chicago, and New York.

Subscription_Length_Months: The number of months the customer has been subscribed.

Monthly_Bill: Monthly bill amount for the customer.

Total_Usage_GB: Total usage in gigabytes.

Churn: A binary indicator (1 or 0) representing whether the customer has churned (1) or not
(0).

5. Data Preprocessing

Data preprocessing was performed to make the dataset suitable for machine learning.
Steps performed:
Converted TotalCharges column from object to numeric format
Handled missing values by removing null records
Converted the target variable Churn into binary values (0 and 1)

Removed customerID column as it is not useful for prediction
Applied one-hot encoding to convert categorical variables into numerical format
These steps improved data quality and ensured better model performance

6. Exploratory Data Analysis (EDA)

The initial step involved exploring the dataset to understand its structure and characteristics.

* The dataset contains information about 7043 customers with 21 variables.
* All variables have the correct data type, and there are no missing values or duplicate records.
* Descriptive statistics were generated for each variable, revealing insights into customer demographics, subscription details, billing, usage, and churn behavior.
* Gender and Location distributions were analysed, indicating the gender and location distribution of the customers.

7.Model Building

A machine learning model was built using Logistic Regression.

Steps:

The dataset was divided into training (80%) and testing (20%) sets

The model was trained using training data

Predictions were made on test data

Why Logistic Regression?

Simple and efficient for classification problems

Easy to interpret results

Works well for binary classification

8. Model Evaluation

The model was evaluated using the following metrics:

Accuracy: Measures overall correctness of the model

Precision: Measures correct positive predictions

Recall: Measures ability to find actual churn cases

F1-score: Balance between precision and recall

Confusion Matrix:

The confusion matrix was used to analyze:

Correct predictions (True Positives & True Negatives)

Incorrect predictions (False Positives & False Negatives)

The model shows good performance with most predictions being correct.

9. Results and Findings

The model achieved an accuracy of approximately (write your value, e.g., 78%)

Important factors affecting churn include:

Contract type

Monthly charges

Tenure

Key findings:

Customers with flexible contracts churn more

Higher monthly charges increase the chances of churn

Long-term customers are less likely to churn

10. Conclusion

This project successfully developed a machine learning model to predict customer churn. The model provides useful insights into customer behavior and helps identify factors that lead to churn.

Such models can assist businesses in improving customer retention strategies and reducing losses.

11. Future Scope

Improve accuracy using advanced models like Random Forest or XGBoost

Perform hyperparameter tuning

Deploy the model as a web application

Use real-time customer data for prediction

12. References

Kaggle – Telco Customer Churn Dataset

Scikit-learn Documentation

Python Libraries (Pandas, NumPy, Matplotlib, Seaborn)